

SESSION 06

# Spread Composition and Grid Breaks

Five decisions govern every spread, and when it's earned, breaking the grid is one of them.

Portfolio as Narrative · Building the Grid, Part II

# Design for Understanding

## Designing for the Reader, Not for Yourself

- A strong portfolio isn't only about what you think matters.
- **It's about** how clearly your story is understood by someone else.
- **Clarity doesn't come from intention alone**, it comes from design.
- To communicate effectively, you need to understand how people see, read, and process visual information.

*“Design is not just what it looks like and feels like. Design is how it works.”*

— Steve Jobs

### WHAT THE READER SEES FIRST

- Dominant image
- Supporting images
- Captions & specs

*Primary → secondary → tertiary: the order a reviewer's eye actually follows, whether or not you designed for it.*

# Learning Objectives

By the end of tonight, you will be able to:



1

**Apply principles** from visual perception and cognitive science to make more intentional layout decisions.



2

**Use hierarchy, grouping, and flow** to organize the two spreads you build out tonight.



3

**Diagnose layout issues** in your own and your classmates' spreads using a structured set of questions.



4

**Revise in real time** to improve clarity, pacing, and how quickly a reviewer understands what they're looking at.

# Eight Reader-Centered Layout Principles

Eight memory tags. Each one compresses a real study in cognitive or perceptual psychology.

| Principle                | Memory Tag              | Helps You...                                   |
|--------------------------|-------------------------|------------------------------------------------|
| 1. Cognitive Load Theory | <b>Declutter</b>        | Focus each spread by reducing distractions     |
| 2. Dual Coding Theory    | <b>Pair It</b>          | Reinforce visuals with concise supporting text |
| 3. Spatial Contiguity    | <b>Keep It Together</b> | Place related images and text near each other  |
| 4. Gestalt Grouping      | <b>Pattern It</b>       | Use grouping, alignment, and similarity        |
| 5. Visual Hierarchy      | <b>Lead the Eye</b>     | Guide attention using scale and placement      |
| 6. Scanning Patterns     | <b>Flow It</b>          | Align layout to natural eye movement           |
| 7. Information Scent     | <b>Signal the Path</b>  | Use labels and structure to aid navigation     |
| 8. Whitespace            | <b>Let It Breathe</b>   | Use space to emphasize and clarify             |

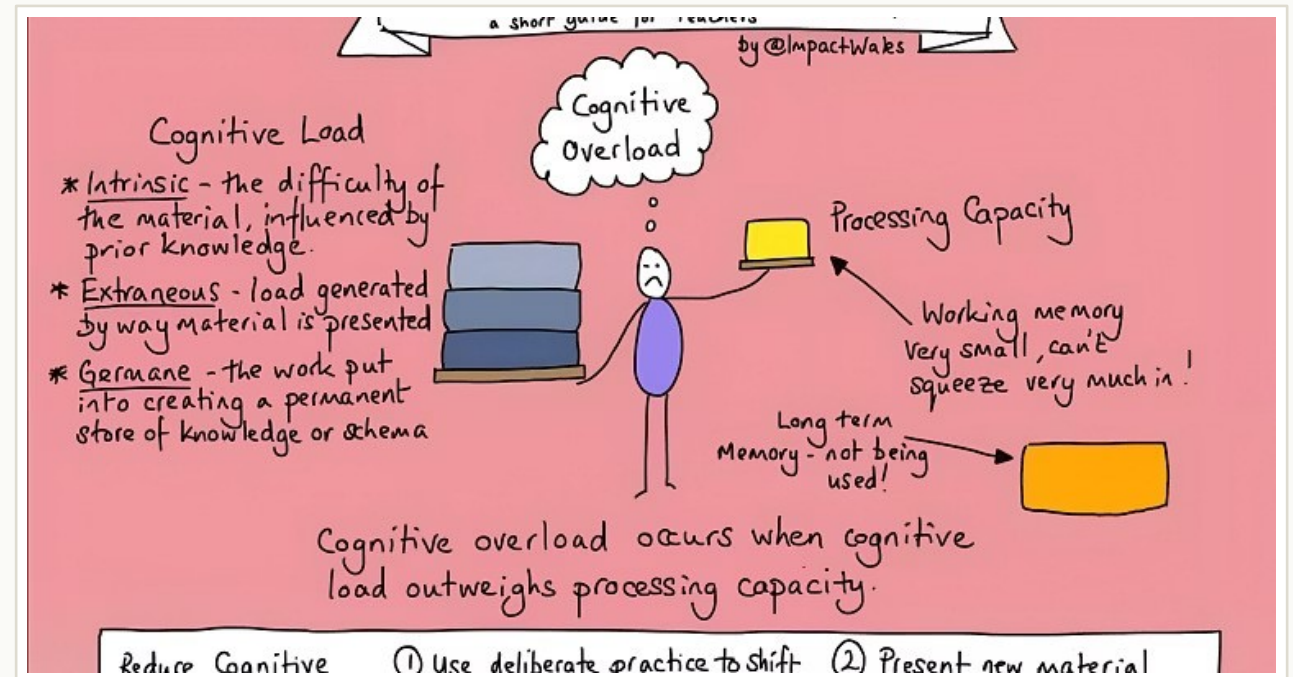
# Principle 1 — Cognitive Load Theory (“Declutter”)

## Theory Summary

Too much information overwhelms the viewer's working memory (Sweller, 1988). Working memory holds roughly four items at a time, for seconds, not minutes. When a spread overloads it, viewers miss key points and fail to retain the content.

## Application

- Focus on one main idea per spread
- Remove visual noise and irrelevant text
- Eliminate decorative elements that don't contribute
- Be ruthless about editing down content



Sweller, J. (1988). Cognitive load during problem solving. *Cognitive Science*, 12(2), 257–285.

# Principle 2 — Dual Coding Theory (“Pair It”)

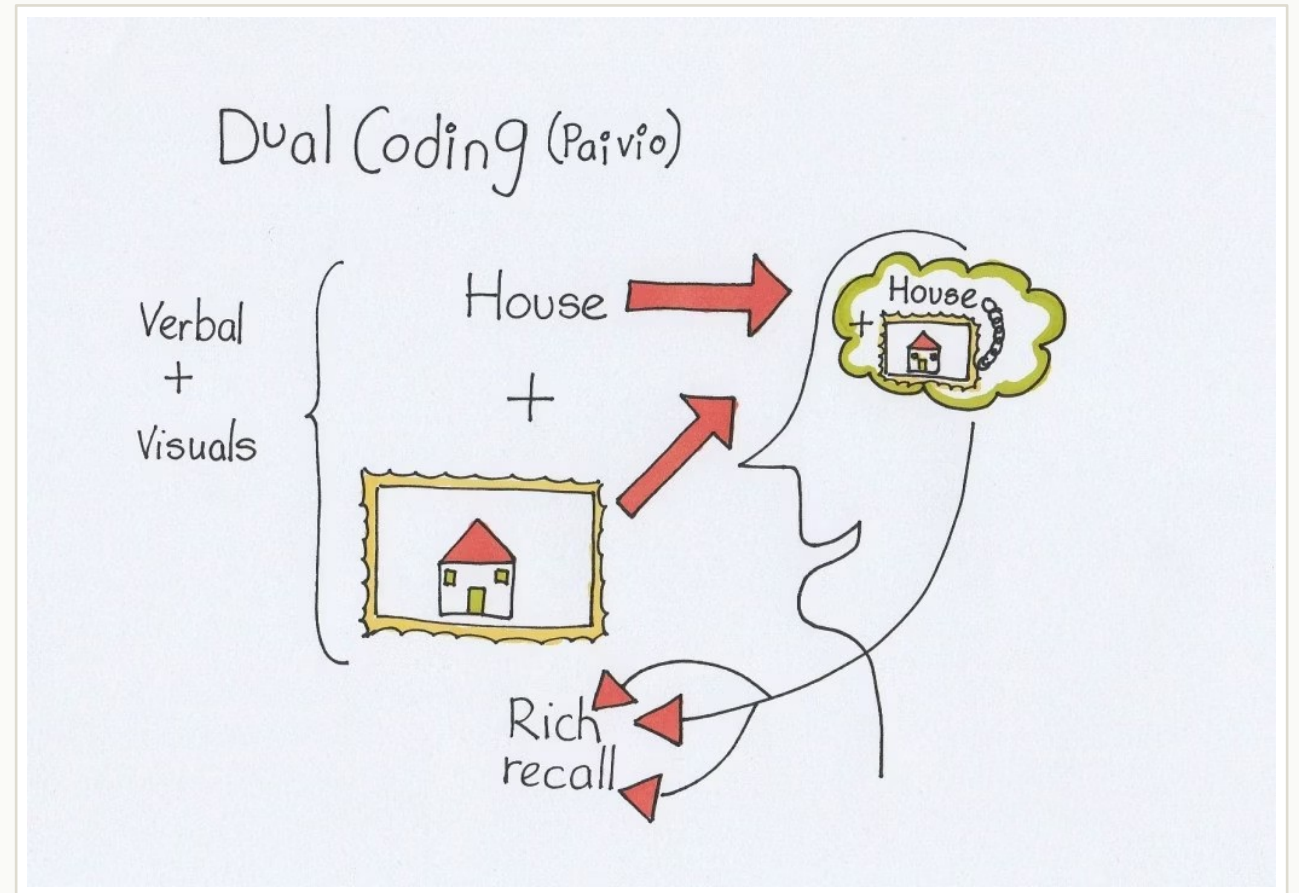
## Theory Summary

We process and retain information better when it's presented through both visual and verbal channels (Paivio, 1971). The brain has separate systems for visual and verbal information. Used together, they reinforce each other.

## Application

- Combine visuals and text to improve memory
- Use concise captions that reinforce, not repeat
- Break up dense text with supporting imagery
- Label diagrams clearly but minimally

Paivio, A. (1971). *Imagery and Verbal Processes*. Holt, Rinehart and Winston.



# Principle 3 — Spatial Contiguity (“Keep It Together”)

## Theory Summary

Learning improves when related words and images are presented near each other rather than separated (Mayer, 2001). The brain spends less effort when it doesn't have to search for and connect related elements.

## Application

- Place labels, diagrams, and captions close to what they describe
- Use directional cues like leader lines only when necessary
- Avoid separating explanatory text from its visual
- Create clear visual connections between related items

## Spatial Contiguity



Place words & pictures near  
each other on the page or screen

*Mayer, R. E. (2001). Multimedia Learning. Cambridge University Press.*

# Principle 4 — Gestalt Grouping (“Pattern It”)

## Theory Summary

Viewers naturally perceive and organize visual elements based on patterns of proximity, similarity, continuity, and closure (Koffka, 1935). The brain automatically groups related elements and seeks order in visual information.

## Application

- **Proximity:** objects near each other appear related
- **Similarity:** similar items are perceived as a group
- **Continuity:** elements aligned in a pattern appear connected
- **Closure:** we mentally complete incomplete shapes

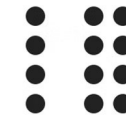
*Koffka, K. (1935). Principles of Gestalt Psychology. Harcourt, Brace and Company.*

## Gestalt Principles



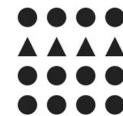
### Good Figure

Objects grouped together tend to be perceived as a single figure. Tendency to simplify.



### Proximity

Objects tend to be grouped together if they are close to each other.



### Similarity

Objects tend to be grouped together if they are similar.



### Continuation

When there is an intersection between two or more objects, people tend to perceive each object as a single uninterrupted object.



### Closure

Visual connection or continuity between sets of elements which do not actually touch each other in a composition.



### Symmetry

The object tend to be perceived as symmetrical shapes that form around their center.

# Principle 5 — Visual Hierarchy (“Lead the Eye”)

## Theory Summary

The brain naturally processes visual information in order of prominence (Lupton, 2004). Deliberate differences in size, weight, and position guide the viewer through your content, primary elements first, tertiary details last.

## Application

- **Primary:** headers and key images command attention first
- **Secondary:** supporting visuals and subheadings provide context
- **Tertiary:** captions and specifications add depth
- Position primary elements at entry points, top or left

*Lupton, E. (2004). Thinking with Type. Princeton Architectural Press.*



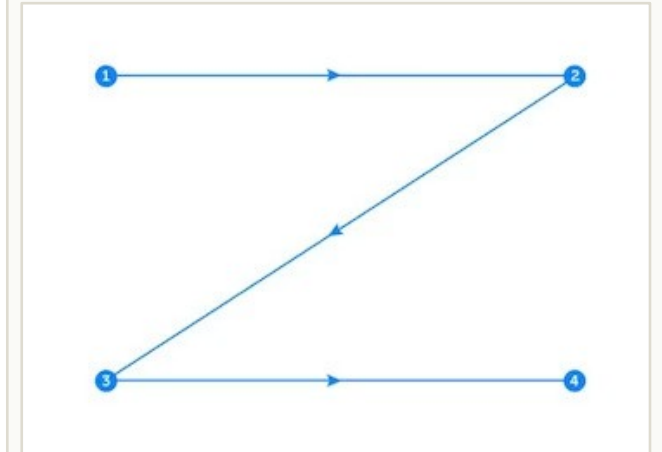
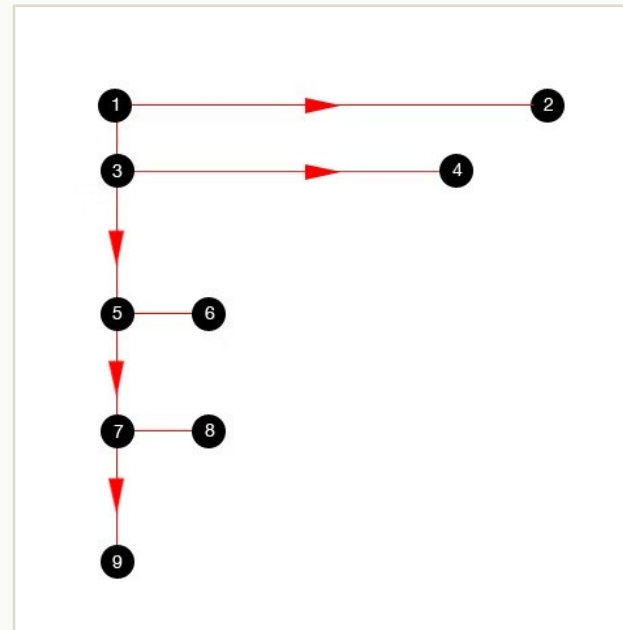
# Principle 6 — Scanning Patterns (“Flow It”)

## Theory Summary

Viewers scan text-heavy pages in an F-shape: across the top, down slightly, across again, then down the left side (Nielsen, 2006). More visual, image-dominant spreads tend toward a Z-path instead. Align your most important content along these natural paths.

## Application

- Position titles, key images, and conclusions at the path's terminal points
- A text-heavy spread: expect an F-path
- A visually dominant spread: expect a Z-path
- Test it: trace where your eye actually lands first



*Nielsen, J. (2006). F-Shaped Pattern for Reading Web Content. Nielsen Norman Group.*

# Principle 7 — Information Scent (“Signal the Path”)

## Theory Summary

Viewers use cues in the environment to predict where to find information and how to navigate content (Pirolli & Card, 1999). Strong “information scent” helps viewers understand content organization and follow your intended narrative.

## Application

- Use consistent, descriptive labels and section headers
- Structure your project as: Concept → Process → Design Development → Result
- Use section markers, color tabs, or page numbers to show progression
- Maintain typographic and layout consistency across projects

Pirolli, P., & Card, S. (1999). *Information foraging. Psychological Review*, 106(4), 643–675.



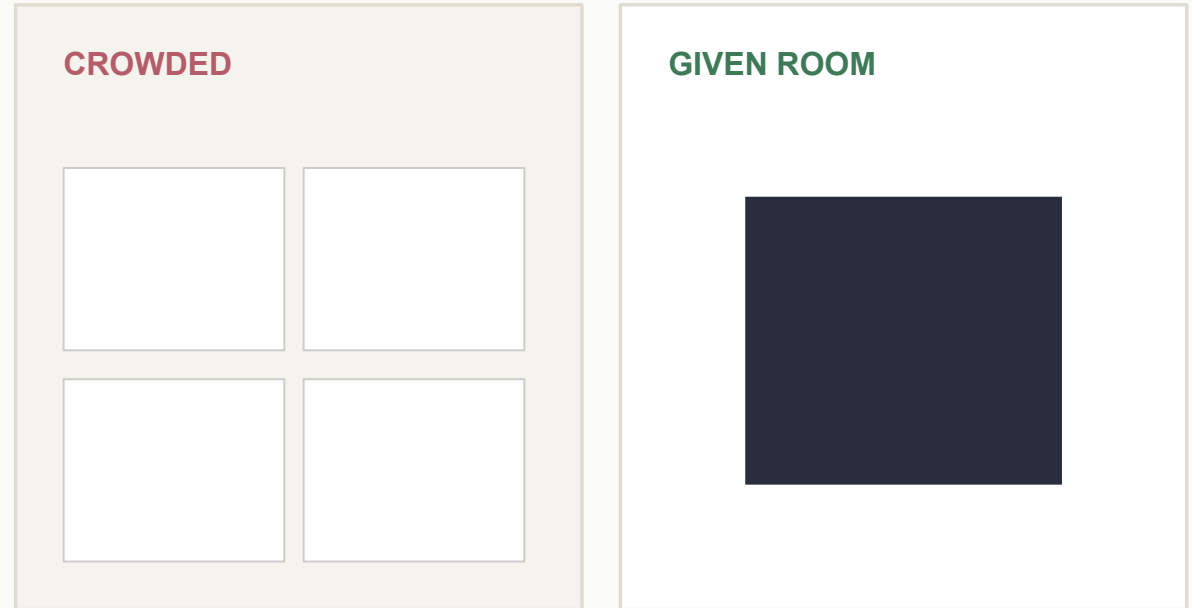
# Principle 8 — Whitespace (“Let It Breathe”)

## Theory Summary

Space is not empty, it's an active design element that supports comprehension and emphasis (Tufte, 1990). The brain needs visual pauses to process information. Whitespace reduces cognitive load and increases perceived value.

## Application

- Use consistent margins around page edges
- Create breathing room between elements
- Allow important elements more surrounding space
- Develop rhythm through alternating dense and open spreads



*The identical drawing, crowded against three neighbors on the left, given generous margins on the right. Nothing about the drawing changed. Its perceived importance did (Tufte, 1990).*

*Tufte, E. R. (1990). Envisioning Information. Graphics Press.*

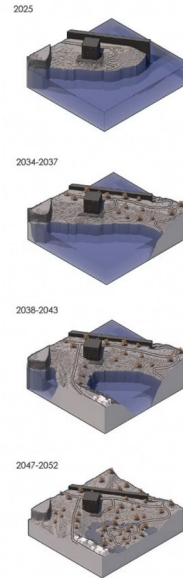
# Principles in Practice

Real student spreads that work, and that don't. Tonight's peer review runs on the same questions.

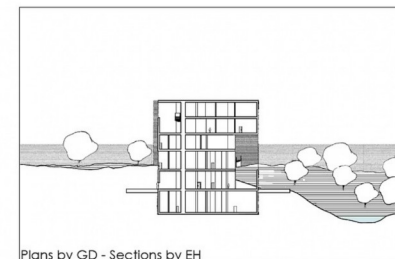
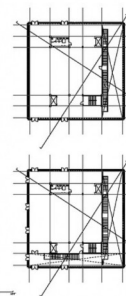
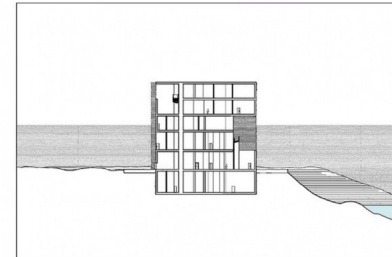
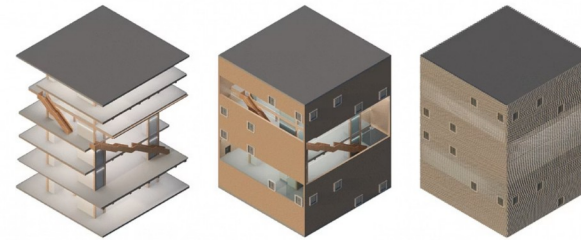
Seven architecture portfolio spreads, student work, evaluated against tonight's eight principles. Three read clearly. Four don't, for reasons you should now be able to name before I do.

## Too Many Dominant Elements, None of Them Win

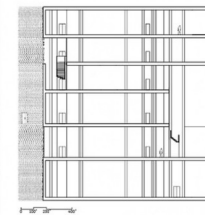
Timeline Maps and Axons in collaboration with GD



Building Layers by GD



Plans by GD - Sections by EH

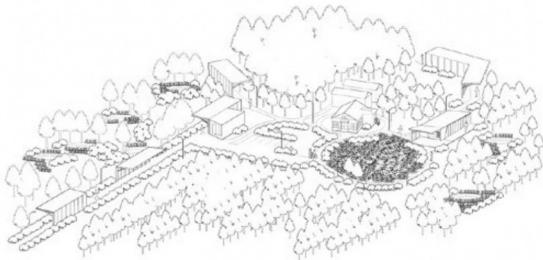
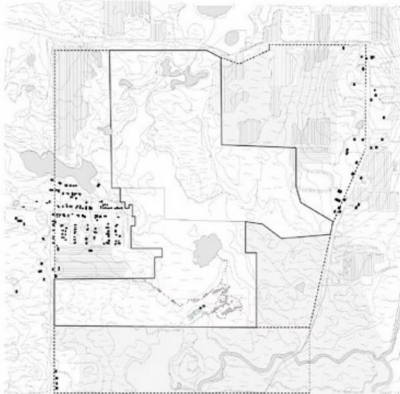


# Dense Paragraphs, No Reading Path



**ABOUT THE PRESERVE**  
Lucia S Nash Nature Preserve is a 650 acre preserve in Burton, Ohio. The Preserve's plentiful ecologies include a kettle lake surrounded by emergent marsh and shrub swamps, the only remaining old-growth boreal white pine forest in Ohio and scattered vernal pools. The preserve is part of a larger 20,000-acre Cuyahoga Wetland complex responsible for providing drinking water for over one million Akron area residents. Snow Lake of Lucia S Nash is the crown jewel of this natural filtration system. The additions bring attention to the relationship between the different ecologies and the important relationship between nature and those experiencing it.

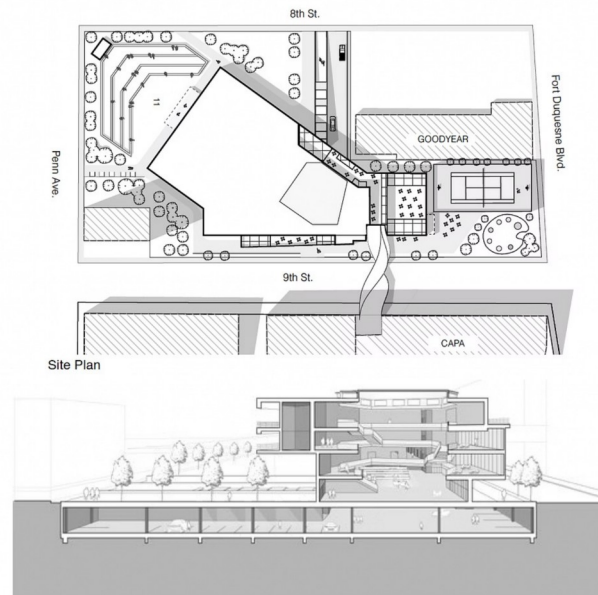
**SITE DESIGN**  
The Site is located in Burton, Ohio of Geauga County. Although the Site is in Ohio where there are 4 distinct seasons, the nature preserve is only open to the public throughout the warmer months of the year. This was a major design driver when designing the expansion. This plan provides a clear depiction of the placement of pavilions, restroom facilities, and residential accommodations in relation to Snow Lake, the expansive marshland, the dispersed vernal pools, and the encompassing woodlands.



**PATH DESIGN**  
The enhancement of pathways serves to seamlessly bridge between the existing elements and the additions, while ensuring a more welcoming connection between the expansion and the preserve's entrance for the general public. This endeavor aims to establish an environment reminiscent of a campus, characterized by an network of pathways and architectural elements guiding visitors toward scenic vistas, natural features, and structures.



# A Full Program Key Instead of a Reading Path

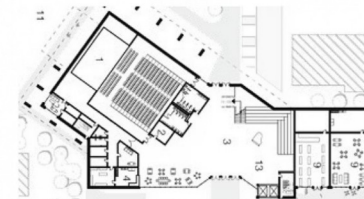


The top image offers a comprehensive view of the site plan, integrating the structure with its urban surroundings and emphasizing its alignment with the cultural district's artistic and performance-oriented vision. These plans introduce outdoor spaces that invite city dwellers to engage with art and nature, providing a vibrant social atmosphere. The perspective below offers a unique insight into the Annex, prominently featuring the expansive cultural lobby and circulation areas that shape social interactions throughout the building. Stretching from the ground floor to the fourth-floor skylight, these spaces infuse the interior with fresh air and natural light, promoting a constant sense of inspiration and creativity for students and visitors.

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3RD FLOOR PLAN



2ND FLOOR PLAN



1ST FLOOR PLAN

These plans meticulously articulate the programmatic layout of the CAPA Annex. The structure unfolds from an underground parking garage, not depicted here. It then ascends to the primary public floor, featuring a 200-seat performance space complete with a comprehensive backstage area, including green rooms and dressing rooms. Adjacent to this, a spacious media lobby serves as a versatile hub, functioning as both a gallery space and a platform for performance art. The lobby also has a cafe for students and visitors and for performance concessions. The third floor is where the CAPA connection bridge connects as well as where CAPA exchange students are housed. The fourth floor similarly consists of housing, but for visiting faculty.

## PROGRAM KEY

- 1 Theater
- 2 Theater Support
- 3 Gallery
- 4 Cafe
- 5 Ensemble/Practice Rooms
- 6 CAPA Connection
- 7 Building Service
- 8 Housing
- 9 Retail
- 10 Parking
- 11 Outdoor Venue
- 12 Gym/Multipurpose

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## Six Drawing Types, One Flat Hierarchy



## One Dominant Image, Everything Else Supports It



*rendering of the southeast plaza. a playground provides space for children to play and seating for families to gather.*

8 | Nick McIntosh



*ground floor plan* ground floor contains programs such as pharmacy, market, workshare, communal living room, maker space, biogarden, communal workout room, cafe, swimming pool, communal garden, basketball court, and playground.

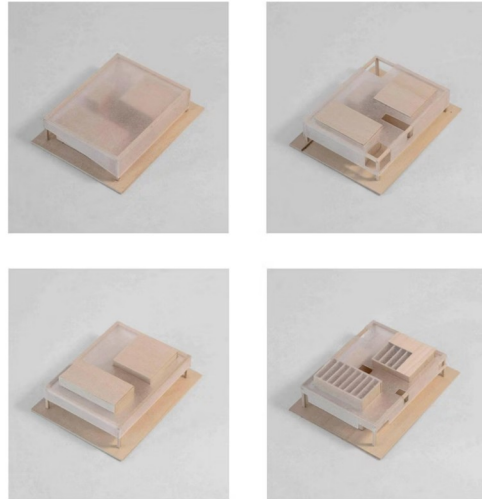


*e - w building section (cutting through sport plaza and playground plaza.)*

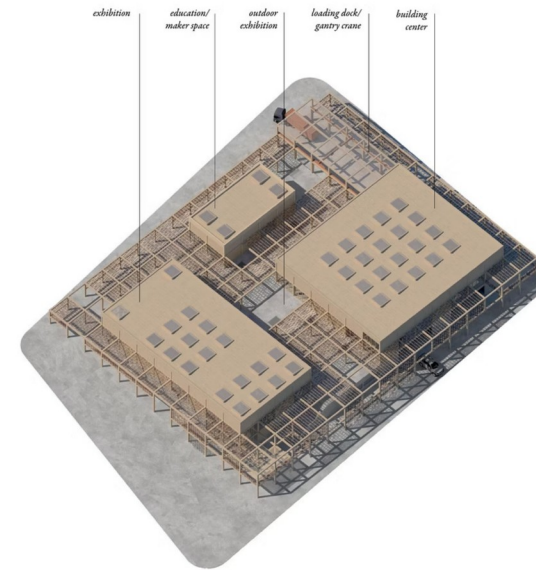
Generative Housing | 9

## Similar Elements, Grouped and Labeled Consistently

*physical model iterations explore possibilities for the relationship between shading screen and program blocks.*



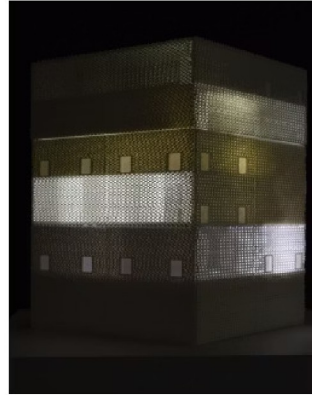
20 | Nick McIntosh



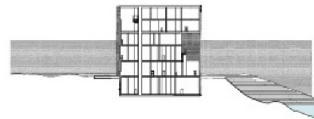
Flexible Framework | 21

## A Full-Bleed Hero, Supporting Detail Scaled Down

Key takeaways from this project include studying how landscape and architecture may collaborate in different environments and potential post-occupancy developments that may take place in original design.

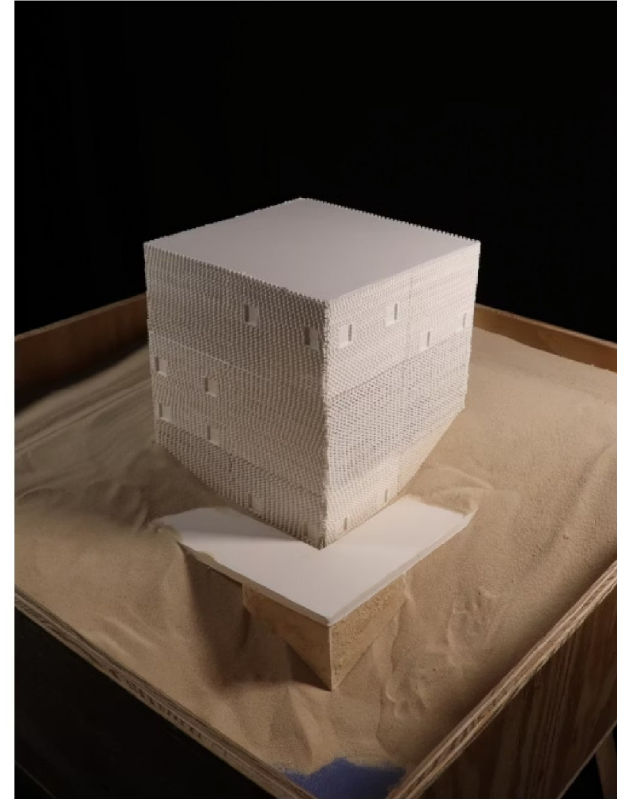
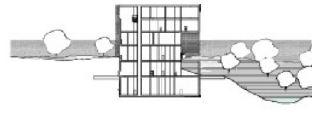


Green Island, 2025  
Section A



16

Green Island, 2025  
Section A



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# From Principles to Decisions

Every reader-centered principle tonight has a direct counterpart in the five spread decisions.

| TONIGHT'S PRINCIPLE                   |   | THE SPREAD DECISION IT DRIVES  |
|---------------------------------------|---|--------------------------------|
| Visual Hierarchy                      | → | <b>Dominance</b>               |
| Scanning Patterns                     | → | <b>Left-Right Relationship</b> |
| Visual Hierarchy (levels)             | → | <b>Image Scaling</b>           |
| Whitespace                            | → | <b>Whitespace</b>              |
| Spatial Contiguity + Gestalt Grouping | → | <b>Composite Logic</b>         |

*The five decisions are not a separate system. They're what the eight principles look like once you're actually placing content on a page.*

# Grid Breaks, Earned

## The Full-Bleed Image

No margins, no columns, no modules. Works as punctuation: a section divider, a project opener, a closing image. Earn it first, don't open with it.

## The Oversized Element

Crosses column or module boundaries to signal hierarchy. Still aligns to column edges, it simply claims more modules than its neighbors.

## The Deliberate Void

A title page with a name, a one-line statement, and one image, using the grid's full capacity by choosing to fill only a fraction of it.

*The one test: can a reviewer see that the break is intentional? If it looks like an accident, it undermines the argument. If nothing on your spread earns a break tonight, leave the grid intact.*

## ACTIVITY

# Tonight's Activity

Open your real project file from Session 05, the one with your grid, baseline, and paragraph styles already built, and start placing real content into it.

- Two spreads, minimum, with real drawings and renderings, not placeholder boxes
- Name the dominant element on each spread before placing anything else
- Captions placed by spatial contiguity, directly against what they describe
- Consistent labels: a section marker, project title, or page number on each
- A grid break only if a spread has actually earned one

## TIMING

**35 minutes**

Then 20 minutes of small-group peer review using tonight's evaluation questions.

## DUE TONIGHT

Your real project file (.indd), two grid-aligned spreads built out with real content, submitted before you leave.

# Design for the Reader, Not the Instinct

Eight principles, five decisions, one test for every grid break. None of it replaces a good project. All of it decides whether a reviewer actually sees one.

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## OPTIONAL READING

“How the Eye Reads a Spread”, the research behind tonight's eight principles.

## NEXT SESSION

Session 07: Typography as Architectural Voice